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Research Article

An Integer Linear Programming Model for the Examination Timetabling Problem: A Case Study of a University in Thailand

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This paper considers the examination timetabling problem (ETTP) at the College of Industrial Technology (CIT) of the King Mongkut's University of Technology North Bangkok in Thailand. A new integer linear programming (ILP) formulation for the ETTP at the CIT is presented. The objectives were to minimize both the number of examination days (the main objective) and the number of rooms used throughout the entire examination period (the secondary objective). In this paper, a course can have multiple sections, and an examination room can accommodate exams for more than one course section. To illustrate the proposed ILP model, real data on courses acquired from the CIT were used to generate two test problems: a small problem and a large problem. The small problem included 32 courses with 69 sections. The large problem included 73 courses with 341 sections, which was the real data required for generating the midterm examination timetable for all first-year courses in the first semester of 2021. Both problems were solved using the CPLEX solver software. The results show that the proposed model could find an optimal examination timetable for the small problem with a computational time of 2 min and 46 s. It also could find a good feasible midterm examination timetable that satisfied the requirements of the CIT for the large problem within the 2-h time limit, much less time than that compared to manual scheduling by the CIT's administrative staff. The obtained midterm examination timetable required five examination days and could reduce 104 examination rooms compared to assigning each course section to a separate examination room. The proposed ILP model can be used in a real-life situation and can be a good option to generate an optimal schedule or a good feasible schedule for examinations at the CIT or other institutions that have similar requirements.

1. Introduction

Timetabling problems are specific types of scheduling problems that involve the assignment of limited resources to a set of activities or events under a set of constraints such that a set of desirable objectives will be met as much as possible [1]. Timetabling problems are attractive and have been widely studied for more than four decades. The educational timetabling problem is a type of timetabling problem that is faced by every educational institution. According to Cataldo et al. [2], the educational timetabling problem is often divided into two classes, which are the course timetabling problem (CTTP) and the examination timetabling problem (ETTP).

The goal of the CTTP is to find a method to allocate the events, such as courses, teachers, and students, to rooms and a fixed number of time slots throughout the five working days of the week [3]. This problem usually occurs at the beginning of each semester at every university. An effective and satisfactory weekly course timetable is the objective for every educational institution because it can increase the utilization of the resources and can fulfill the preferences of teachers, students, and the people who handle the CTTP. However, the requirements and satisfactions of the course timetable can differ from institution to institution due to the individual specifications of the institution.

At the middle or end of each semester, each university always has to deal with the midterm or final examinations.

This process can be considered as the ETTP. The ETTP involves assigning exams to a given number of time slots within the examination period while satisfying various required constraints [4]. Note that the ETTP faced by universities can also differ from one institution to another because each university may need to address some unique restrictions depending on that particular institution's rules.

The constraints in the educational timetabling problem can be categorized into two types: hard constraints and soft constraints [3]. Hard constraints are those that must be satisfied without any violation. If a hard constraint is not satisfied, the timetable is not feasible for the problem. On the other hand, soft constraints are the constraints that are desirable but not necessary to be satisfied. These can be violated at the cost of contributing a penalty to the objective function. The more soft the constraints that are satisfied, the higher the quality of the solution for the timetable. An example of a hard constraint in the CTTP is that an instructor must be assigned to only one course at any given time. An example of a soft constraint in the CTTP is that a classroom should be appropriately sized to accommodate the number of students. In the ETTP, an example of a hard constraint is that each exam has to be scheduled to a time slot. An example of a soft constraint in the ETTP is to spread the exams throughout the examination period so that the students have enough free time for revision between exams. Due to the unique requirements and rules of each institution, a given constraint could be a hard constraint in one institution but a soft constraint in another.

The CTTP has been fairly widely studied. Dimopoulou and Miliotis [5] proposed an integer linear programming (ILP) model for the CTTP and solved the problem using the data from the Athens University of Economics and Business. Daskalaki et al. [6] reported on the application of an ILP model to the CTTP and tested it on the data from the Department of Electrical and Computer Engineering at the University of Patras. In Schimmelpfeng and Helber [7], the CTTP was investigated as a case study of the School of Economics and Management at Hannover University, and the authors constructed the complete course timetable using an ILP approach. Prabodanie [8] presented an ILP model for the CTTP to find an optimal course timetable at the Faculty of Applied Sciences, Wayamba University of Sri Lanka. Recently, Zaulir et al. [9] proposed an ILP model for the CTTP that included many essential constraints commonly used in most universities. The model was applied to find an optimized course timetable using the data from a university in Malaysia. Apart from the mathematical programming model, several efficient metaheuristic approaches have also been developed for solving the CTTP, such as genetic algorithm (GA) [10, 11], simulated annealing (SA) [12], tabu search [13, 14], particle swarm optimization (PSO) [15], and ant colony optimization [16]. Moreover, hybrid metaheuristics have also been developed to solve the CTTP. For example, Akkan and Gulcu [17] developed hybrid metaheuristics consisting of GA, SA, and hill climbing to solve the CTTP. In addition, the hybrids of GA and tabu search are also applied to solve the CTTP in Wong et al.'s study [18]. A more exhaustive review of solution methodologies for solving the CTTP also can be found in Abdipoor et al. [19].

According to Bassimir and Wanka [20], the ETTP can be classified into the two types of postenrollment ETTPs (PE-ETTPs) and curriculum-based ETTP (CB-ETTP). In the PE-ETTP, the number of students registered for the exams is known before the timetable is generated, and it is known that which exams should not be held at the same time to avoid exam conflicts for individual students. The CB-ETTP is the pre-enrollment case of the ETTP, where the number of students registered for courses is not known, but the timetable is generated based on the curriculum configurations. Thus, in the CB-ETTP, the examination timetable can be announced at the beginning of the semester and created in advance by assigning a day, a time, and a classroom to the examination of each course using the estimated number of students taking an exam.

There are many research studies on the ETTP in the literature in which diverse mathematical programming and metaheuristic approaches have been developed for solving a variety of the ETTPs. MirHassani [1] proposed an ILP model for the PE-ETTP, where the model considered the maximization of paper spread in the examination timetable and was tested on the real data from the Shahrood University of Technology. Al-Yakoob et al. [21] presented an ILP model for the PE-ETTP at Kuwait University that considered many features, such as the proctors' preferences, traffic considerations, and certain imposed gender policies. In Bergmann et al. [22], an ILP model for the PE-ETTP was presented where the model considered the workload associated with an exam, the need for splitting a large exam over several rooms, and the assignment of exams to inconvenient periods. The model was tested with the data from the Humburg University of Technology in Germany. Cataldo et al. [2] presented an ILP model for solving the CB-ETTP using real-world instances provided by the Faculty of Engineering at the Universidad Diego Portales in Santiago of Chile. Furthermore, Erdos and Kovari [23] proposed a mixed ILP (MILP) model for scheduling the final oral exam, where only one student at a time can take the oral exam in front of a board of instructors in a room. The proposed model was solved using the data from the Budapest University of Technology and Economics.

In Al-Matouq [24], a new convex optimization framework for solving the ETTP that can be described as ILP is presented. The method is based on converting the ILP into a cardinality constrained problem while relaxing the operational constraints. The problem was solved using weighted least absolution shrinkage and selection operator (LASSO) with weights that were updated using linear programming to lead the solution toward a binary solution. The proposed method was used to solve benchmark timetabling problem from the 2007 International Timetabling Competition and could improve the best documented result in the literature. Furthermore, Mauritsius et al. [25] considered the ETTP as a graph coloring problem and proposed iterative SA heuristics for solving the problem using the Toronto benchmark dataset introduced by Carter et al. [26]. The results showed that their method is capable of finding the best solutions ever reported as well as improving current best-known results. In addition, Carlsson et al. [27] presented mixed integer programming, constraint programming, and SA for solving the ETTP using the benchmark instances based on real data from various Italian universities. The results showed that their approaches outperformed other approaches in the literature on the majority of instances. Other metaheuristics, moreover, have been developed for solving the ETTP, such as GA [28], SA [29], PSO [30], bee colony optimization [31], and intelligent water drops algorithm [32]. Besides, hybrid metaheuristics have also been developed to solve the ETTP, such as hybrids of tabu search and GA [33]. A more exhaustive review of solution methodologies for solving the ETTP can be found in Siew et al. [34].

This paper considers a real-world ETTP application. We investigate a case study of the College of Industrial Technology (CIT) of the King Mongkut's University of Technology North Bangkok (KMUTNB). Currently, the KMUTNB has 11 faculties, and the CIT is one of them (the CIT is equivalent to a faculty-level unit in the KMUTNB). At the CIT, the examination timetable is scheduled manually by the administrative staff. They usually construct the current examination timetable by replicating the timetable of the previous year with minor changes. In recent years, however, changes may occur more frequently, and revising what has been developed historically is not always the best approach. Furthermore, constructing the examination timetable manually is not suitable because it is time-consuming, tedious, and difficult. In addition, the manual timetable can be inefficient and is often accompanied by shortcomings, such as exam conflicts. Generating a conflict-free examination timetable is difficult and so it would be better if the timetable was generated via proper modeling or analytical tools. Therefore, this paper aimed to develop an automated procedure for constructing an efficient and effective examination timetable for the CIT. To this end, we focused on an ILP method, which is a type of mathematical programming model that can explicitly describe the objective, constraints, and all the other characteristics of the scheduling problems [35] and so can include the timetabling problems. It is also usually the first natural way to solve scheduling problems and can provide an optimal solution if one exists [36]. The developed ILP model for the ETTP at the CIT is presented in this research study. It is a new application for the CIT because, to the best of our knowledge, an ILP model for constructing an examination timetable for the CIT has never been investigated in the literature. The proposed ILP model can close the gap between theory and practice in the examination timetabling and can be used as an option to automatically create an examination timetable for the CIT, which is the main contribution of this paper.

The remainder of this paper is organized as follows. In Section 2, the problem description of the ETTP at the CIT is presented and then the proposed ILP model is presented in Section 3. The obtained numerical results are shown in Section 4. Finally, the discussion and conclusion are drawn in Sections 5 and 6, respectively.

2. Problem Description

This paper outlines the development of a mathematical programming model for the ETTP at the CIT of the KMUTNB in Thailand. The problem description of the ETTP at the CIT is detailed in this section. At the KMUTNB, the academic year of every faculty is divided into two semesters. There are two normal examinations in each semester, which consist of midterm and final examinations. In the KMUTNB, both midterm and final examinations are organized by the respective faculty independently. Since the CIT is a faculty within the KMUTNB, the CIT organizes both midterm and final examinations for their own courses by themselves.

At present, the CIT offers a total of 30 bachelor's degree curriculums, with 22 being regular curriculums (for students who graduated from high school) and eight being continuous and linking curriculums (for students who graduated with a high vocational certificate from a community college). Several courses are taught in each semester at the CIT for students from all curriculums, and a given course can be common to students from different curriculums. Moreover, a course may be divided into several sections. Each section is taught in a single classroom by an instructor, but all sections cover the same course content. Each section of a course is fixed for enrollment by students from a specific curriculum. An example is illustrated in Figure 1. In this figure, the CIT offers a total of 30 bachelor's degree curriculums. Suppose that the total number of courses taught in the current semester at the CIT for students from all curriculums is s, where the number of sections of each course $i \in \{1, 2, \ldots, s\}$ is g_i. Assume that Course 1 is "Tools Engineering," which has three sections. This course will be taken by students from two curriculums, i.e., Curriculums 1 and 2. Students in Curriculum 1 are divided into two sections for Sections 1 and 2 of Course 1, while students in Curriculum 2 are assigned to study Course 1 in Section 3. Note that for each course, the CIT has already provided the information that students in which curriculum have to study in which section of that course.

The scope of this paper is that we aim to schedule the midterm examination for the overall courses of the first-year bachelor's degree students at the CIT. From now on, therefore, courses and students in this paper mean the first-year undergraduate student courses and the first-year undergraduate students, respectively. More details of the ETTP at the CIT can be described as follows:

- The number of all available courses in the current semester that have to be set up for the examination is known.
- A course can have more than one section, and the number of sections of each course is given.
- For each course, the number of registered students in each section is known.
- The list of courses that must be studied by the students in each curriculum in the current semester is provided (i.e., it is known which courses in the current semester that the students in each curriculum have to study).

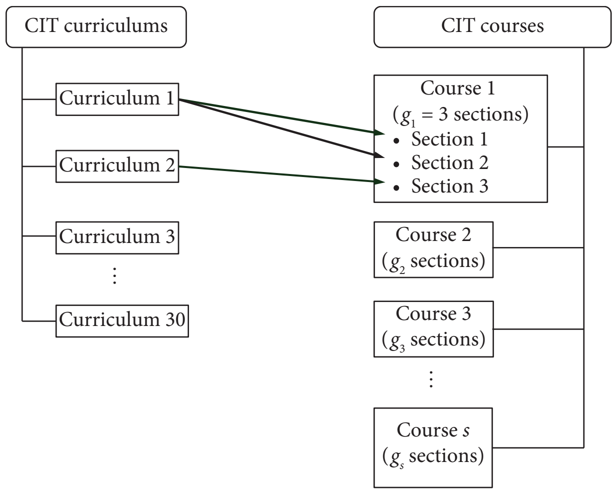

FIGURE 1: An example of student enrollments in a course at the CIT.

Note that most courses in the study plan in the study bulletin for the first-year students are mandatory. The students usually study these courses during their first year, and these courses will be scheduled for the examinations by the faculty.

- In the middle of the semester, there is a 6-day period of midterm exams starting from Monday to Saturday (it does not include Sunday).
- On every examination day, there are two different intraday time slots for examinations, that is, 9.00–12.00 a.m. and 1.00–4.00 p.m. Thus, the total number of time slots for the midterm examination at the CIT is 12 (two per day for six days).
- The number of rooms and the number of seats (capacity) in each room provided for the examination are known.
- In the examination at the CIT, all sections of a course must be scheduled to have their exams at the same time.
- All students in the same section of a course must be scheduled to take the exam in the same room; that is, they cannot be split into two separate examination rooms.
- Each room can only be used for the examination of at most one course at a time; that is, any two different courses cannot be assigned to the same room for the examination. The CIT needs this condition to prevent students from taking exams for the wrong courses, which can lead to various issues, such as an insufficient number of exam papers in the examination room (the exam papers are wasted due to students taking the wrong courses) or the need to relocate students during the exam. Furthermore, some courses cannot be scheduled to have their exams in the same room due to differing exam conditions. For example, a course allows a calculator to be brought into the examination room, while another course does not permit it, or a course exam is an open book, while another course exam is not.
- An examination room can have exams for more than one section of the same course at a time. For example, an examination room can accommodate all three sections of Course 1 (Tools Engineering) if the total number of students from all three sections does not exceed the capacity of this room.

Moreover, the following assumptions are made in this paper:

1. All students in a curriculum will study all the courses according to the study plan in the study bulletin of the current semester. For example, assume that the courses corresponded to the study plan of Curriculum 1 in the current semester consist of Tools Engineering (Tool Eng), Calculus (Cal), Computer (Com), Chemistry (Chem), and Physics (Phy). Then, all the students in this curriculum study these five courses in the current semester and so to avoid exam conflicts, these five courses cannot be scheduled to have their exams at the same time. Note that an exam conflict occurs if two exams are scheduled at the same time slot and there is at least one student that has to take both exams. A timetable is said to be conflict-free for every student if no student has to take more than one exam at the same time. The data of courses in the study plan in the study bulletin of each curriculum were used as the condition to avoid an exam conflict.
2. This paper assumes that each course at the CIT requires the same duration for the examination, which is 3 h.

There are two objectives (soft constraints) in scheduling the examination that are considered at the CIT as follows:

1. The first objective is the main objective and is to schedule the exam to minimize the number of examination days used over the entire examination. The fewer the examination days the CIT uses, the more it can save on costs, as there will be no need for rooms, buildings, and staff resources on the remaining nonexam days of the examination period.
2. The secondary objective is to minimize the number of rooms used for the examinations of all courses over the entire examination. The fewer the examination rooms the CIT uses, the more efficiently the CIT utilizes its resources.

Both objectives can reduce the cost of the examinations, such as the electricity consumption and the cost of proctors. In this paper, since the number of students participating in a given course exam is known before the examination timetable is generated and the list of courses studied by each individual student is known to avoid exam conflict, the ETTP at the CIT considered in this paper can be categorized as a PE-ETTP.

3. Proposed Mathematical Model

This section presents the ILP model for the ETTP at the CIT, as described in Section 2. The parameters, indices, and variables are defined below.

3.1. Indices

i. i is the index of courses.
ii. j is the index of sections of a course.
iii. k is the index of time slots.
iv. l is the index of examination rooms.
v. h is the index of curriculums of the CIT.

3.2. Parameters

i. s is the number of courses to be scheduled for the exams, where $i \in S = \{1, 2, \ldots, s\}$ and S represents the set of all courses.
ii. g_i is the number of sections of course i, where $j \in G_i = \{1, 2, \ldots, g_i\}$ and G_i represents the set of all sections of course i.
iii. m_{ij} is the number of students registered in section $j \in G_i$ of course $i \in S$.
iv. t is the number of all time slots in the examination period, where $k \in T = \{1, 2, \ldots, t\}$ and T represents the set of all number of available time slots. In our case, $t = 12$.
v. r is the number of examination rooms, where $l \in R = \{1, 2, \ldots, r\}$ and R represents the set of all available rooms.
vi. c_l is the number of seats or capacity of room l.
vii. n is the number of all bachelor's degree curriculums at the CIT, where $h \in N = \{1, 2, \ldots, n\}$ and N represents the set of all curriculums.
viii. A_h is the set of all courses that must be studied by the students in curriculum h according to the study plan in the study bulletin in the current semester. For example, if $A_1 = \{\text{Tool Eng, Cal, Com, Chem, Phy}\}$, it means that every student in Curriculum 1 studies Tool Eng, Cal, Com, Chem, and Phy in this semester according to the assumption in Section 2.
ix. L is a large positive number.
x. ε is an infinitesimally small positive number.

3.3. Decision Variables

i. y_{ik} 1, if course i is assigned to time slot k for the examination, and 0 otherwise.
ii. x_{ijl} 1, if the section j of course i is assigned to room l for the examination, and 0 otherwise.
iii. z_{ikl} 1, if room l is used for the examination of course i in time slot k, and 0 otherwise.
iv. d is the integer variable representing the number of required examination days to cover all the exams.

In the proposed model, the variable z_{ikl} is defined to count the number of rooms required for the examination of course i. Actually, all three decision variables are related to each other. Note that if $x_{ijl} = 1$ and $y_{ik} = 1$, these mean that section j of course i is assigned to room l in time slot k for the examination and then the variable z_{ikl} must be equal to 1 because room l is used for the examination of course i (section j) in time slot k. In addition, it is possible that there is more than one section of the same course that is assigned to the same room in the same time slot for the examination. For example, if sections j and j' of course i are assigned to the same room l in time slot k for the examination, that is, $x_{ijl} = x_{ij'l} = 1$ and $y_{ik} = 1$. In this case, the variable z_{ikl} must also be equal to 1, which means that room l is counted as having been used only once even if both sections j and j' of course i are assigned to this room in time slot k for the examination.

The ILP model for the ETTP at the CIT can be stated as follows:

$$\min\left(d + \varepsilon \sum_{i=1}^{s}\sum_{k=1}^{t}\sum_{l=1}^{r} z_{ikl}\right), \tag{1}$$

s.t.

$$\sum_{k=1}^{t} y_{ik} = 1 \quad \forall i \in S, \tag{2}$$

$$\sum_{i \in A_h} y_{ik} \le 1 \quad \forall k \in T, \forall h \in N, \tag{3}$$

$$\sum_{l=1}^{r} x_{ijl} = 1 \quad \forall i \in S, \forall j \in G_i, \tag{4}$$

$$m_{ij} - c_l \le \left(1 - x_{ijl}\right)L \quad \forall i \in S, \forall j \in G_i, \forall l \in R, \tag{5}$$

$$x_{ijl} + y_{ik} \le 1 + z_{ikl} \quad \forall i \in S, \forall j \in G_i, \forall k \in T, \forall l \in R, \tag{6}$$

$$\sum_{i=1}^{s} z_{ikl} \le 1 \quad \forall k \in T, \forall l \in R, \tag{7}$$

$$\sum_{j=1}^{g_i} m_{ij}x_{ijl} \le c_l \quad \forall i \in S, \forall l \in R, \tag{8}$$

$$k y_{ik} \le 2d \quad \forall i \in S, \forall k \in T, \tag{9}$$

and

$$\begin{aligned} y_{ik} &\in \{0,1\} \quad \forall i \in S, \forall k \in T, \\ x_{ijl} &\in \{0,1\} \quad \forall i \in S, \forall j \in G_i, \forall l \in R, \\ z_{ikl} &\in \{0,1\} \quad \forall i \in S, \forall k \in T, \forall l \in R, \end{aligned} \tag{10}$$

where d is an integer.

Objective function (1) consists of two terms. The first term (the main objective) is to minimize the number of required examination days to cover all the exams. The second term in the objective function (the secondary objective) is to minimize the number of rooms required for the examinations of all courses over the entire examination schedule, which means that the number of rooms assigned to each course for the examination is minimized. Since the first term of minimizing the number of days to cover all the exams is the main objective, the first term must be given the importance to be firstly minimized before the second term. Therefore, the constant ε in the second term is set to be sufficiently small so that the first term is prioritized for minimization before the second term.

Constraint (2) ensures that each course must be assigned to a time slot for the examination. Note that if course i is assigned to time slot k, it means that all sections of course i are assigned to time slot k for examinations; that is, the examinations of all sections of a course occur at the same time.

Constraint (3) enforces that any two courses in the same curriculum cannot be scheduled for examinations in the same time slot. This constraint guarantees that an exam conflict cannot occur because every student in a curriculum cannot take more than one exam in the courses of his/her curriculum at the same time.

Constraint (4) makes sure that each section of each course must be assigned to a room for the examination. Note that if it is known that section j of course i is assigned to room l for the examination $(x_{ijl} = 1)$, then the time slot for the examination of this course section can be known from variable y_{ik} that is equal to 1.

Constraint (5) requires that if the number of students in a course section is greater than the number of seats in an examination room, then this course section cannot be assigned to this room for the examination. This is because if the number of students of section j of course i (m_{ij}) is greater than the capacity of room l (c_l), then the left-hand side (LHS) of Constraint (5) is positive, and it forces the value of x_{ijl} in the right-hand side (RHS) to be 0. This means that section j of course i cannot be assigned to room l for the examination. On the other hand, if $m_{ij} \leq c_l$, then the LHS is less than or equal to zero. Therefore, the value of x_{ijl} in the RHS can be 1, which means that section j of course i can be assigned to room l for the examination. Note that, from this constraint, it is possible that more than one section of the same course can be assigned to the same room for the examination. For example, if we have sections j and j' of course i and room l such that $m_{ij} \leq c_l$ and $m_{ij'} \leq c_l$, then both variables x_{ijl} and $x_{ij'l}$ can be equal to 1 according to Constraint (5).

Constraint (6) ensures that if a section of a course is assigned to a room in a time slot, then this room is counted as used for the examination of that course in that time slot. This is because if section j of course i is assigned to room l in time slot k for the examination $(x_{ijl} = 1$ and $y_{ik} = 1$, or the LHS is equal to 2), then the variable z_{ikl} in the RHS will be equal to 1. On the other hand, if section j of course i is not assigned to room $(x_{ijl} = 0)$ or course i is not assigned to time slot k $(y_{ik} = 0)$ for the examination (the LHS is less than or equal to 1), then variable z_{ikl} in the RHS will be equal to 0 due to the minimization of the second term in the objective function. Furthermore, according to Constraint (6), room l will be counted as having been used only once even if more than one section of the same course is assigned to room l in a time slot for the examination. For example, assume that sections j and j' of course i are assigned to the same room l in time slot k for the examination, that is, $x_{ijl} = x_{ij'l} = 1$ and $y_{ik} = 1$. Then, we have the constraints $x_{ijl} + y_{ik} \leq 1 + z_{ikl}$ and $x_{ij'l} + y_{ik} \leq 1 + z_{ikl}$. Both constraints enforce that $z_{ikl} = 1$, which means that room l is counted as having been used only once (for the examination of course i in time slot k) as desired.

Constraint (7) enforces that each room in any time slot can be used for the examination of at most one course. This constraint ensures that any two courses that are assigned to the same time slot cannot be scheduled to have their exams in the same room.

Constraint (8) makes sure that if there is a course that has more than one section assigned to the same room for the examination, then the total number of students from all these sections must not exceed the capacity of the room.

Constraint (9) determines the number of required examination days to cover all the exams (d).

The solution to the proposed ILP model provides information about which section of which course should be assigned to which room and at which time slot for the examination. This information could then be used to create the examination schedule for the courses of the students at the CIT, which is useful and practical in the real situation.

4. Numerical Results

To demonstrate the proposed ILP model, we used real-world data from the CIT to generate two test problems for testing the model. The details of the test problems (the small and large problems) and their computational results are shown in Sections 4.1 and 4.2, respectively.

4.1. Test Problems. The two test problems evaluated were a small problem (Problem 1) and a large problem (Problem 2). The data in each problem are the real data on courses, sections, and the number of registered students in each course section from the first semester of the academic year 2021 at the CIT. The details of Problems 1 and 2 are given in Subsections 4.1.1 and 4.1.2, respectively.

4.1.1. Problem 1 (Small Problem). In this problem, we used a small real dataset to test the proposed ILP model for generating an examination schedule. The data in Problem 1 included information on courses, sections, and the number of registered students in each course section from eight continuous and linking curriculums at the CIT $(n = 8)$. For simplicity, these curriculums are called Curriculums 1–8, respectively. The number of courses to be scheduled (s) in this problem is 32, where the number of sections of each course $i \in S$ (or g_i) is shown in Table 1. The number of students in each section $j \in G_i$ of each course $i \in S$ (or m_{ij}) is also shown in Table 1. Note that $S = \{1, 2, \ldots, s\}$ and $G_i = \{1, 2, \ldots g_i\}$. The number of time slots is 12. The number of available examination rooms (r) in this problem is set to be 17, and the number of seats or capacity (c_l where $l \in R = \{1, 2, \ldots r\}$) of each room is shown in Table 2. The set of all courses in each curriculum (A_h where $h \in N = \{1, 2, \ldots, n\}$) is shown in Table 3. The total number of sections from all courses in Problem 1 is 69.

4.1.2. Problem 2 (Large Problem). This problem used data on courses, sections, and the number of registered students in each course section across all the curriculums at the CIT to generate the midterm examination schedule for all first-year students. The data in this problem included (i) 73 courses to be scheduled for the midterm exam; (ii) the number of sections of each course, and the number of students in each section of each course, as shown in Table 4; (iii) 12 time slots (6 days of the midterm examination period); (iv) 66 available examination rooms at the CIT and the capacity of each room, as shown in Table 5; and (v) 30 curriculums at the CIT and the set of all courses in each curriculum, as shown in Table 6. In Problem 2, the total number of sections from all courses is 341.

4.2. Computational Results. In this section, both Problems 1 and 2 were solved using the proposed ILP model. There were two parameters in the proposed model, ε and L. In both problems, the parameter ε was set to be 0.001 to ensure that it was sufficiently small that the first term in the objective function (the number of days needed to cover all the exams: d) is prioritized for minimization. Since the second term in the objective function ($\sum_{i=1}^{s}\sum_{k=1}^{t}\sum_{l=1}^{r} z_{ikl}$) is the number of rooms required for the examination of all sections from all courses, this value cannot be greater than the total number of sections from all courses (the maximum value is 341 sections in Problem 2) when each section uses one room for the examination. When the value of ε is set to 0.001, the maximum value of the second term in the objective function is 0.341. The model will first prioritize minimizing the value of d because d is an integer, but the value of the second term in the objective function is three decimal places, which is less than one. In both problems, therefore, setting ε to 0.001 is sufficiently small. In addition, the parameter L was set to be 200 (the maximum capacity from all available rooms) in both problems. Note that the constant L appears in Constraint (5) in the proposed model, and it is easy to see that the LHS of Constraint (5) is less than $\max_{l \in R} c_l$, which is equal to 200 (c_{17} in Problem 1 or c_{61} in Problem 2). Therefore, the parameter L could be set to 200.

4.2.1. Results of Problem 1. In this subsection, the proposed ILP model was used to solve Problem 1. The model was implemented in the ILOG OPL CPLEX 12.6 software running on a computer with an AMD RADEON R5 2.50 GHz CPU and 8 GB RAM. The model size of Problem 1, which included the number of binary variables, integer variables, and constraints, is shown in Table 7. The model could be solved to an optimal solution for Problem 1 in a computational time of 2 min and 46 s. The results returned values of x_{ijl}, y_{ik}, and z_{ikl} that were equal to 1, and these values could be interpreted to the examination schedule, as shown in Table 8.

Table 1: The number of courses, sections, and students in each section for Problem 1.

Course $i \in S$	**Number of sections of course i (g_i)**	**Number of students in each section ($m_{i1}, m_{i2}, \ldots, m_{i,g_i}$)**
1	6	58, 54, 57, 50, 50, 50
2	4	50, 50, 50, 50
3	2	50, 50
4	1	50
5	1	50
6	1	50
7	5	46, 36, 39, 50, 50
8	1	50
9	1	50
10	1	50
11	2	50, 50
12	2	50, 50
13	5	35, 32, 50, 50, 50
14	1	50
15	2	50, 50
16	2	50, 50
17	2	50, 50
18	3	50, 50, 50
19	2	50, 50
20	2	50, 50
21	2	50, 50
22	1	50
23	1	50
24	3	50, 50, 50
25	3	50, 50, 50
26	5	50, 50, 50, 50, 50
27	1	50
28	1	50
29	1	50
30	1	50
31	3	50, 50, 50
32	1	50

Table 2: The capacity of each room for Problem 1.

Room number ($l \in R$)	**Capacity (c_l)**
1	100
2	100
3	100
4	100
5	80
6	50
7	50
8	50
9	50
10	50
11	50
12	50
13	50
14	70
15	70
16	70
17	200

As shown in Table 8, the examination schedule of Problem 1 used eight time slots to cover all the exams ($d = 4$). The columns in Table 8 represent the time slots of

Table 3: List of courses in each curriculum for Problem 1.

Curriculum $h \in N$	**Courses in curriculum h (A_h)**
1	{1, 2, 3, 4, 5}
2	{1, 2, 3, 4, 6}
3	{1, 2, 4, 5, 6}
4	{7, 8, 9, 10, 11, 12, 13, 14}
5	{2, 15, 16, 17, 18, 19, 20}
6	{2, 21, 22, 23, 24, 25, 26}
7	{2, 24, 25, 26, 27, 28, 29}
8	{2, 22, 24, 25, 26, 30, 31, 32}

each examination day, and each row in the table represents the examination room number. Each cell in Table 8 consists of two numbers. The first number represents the course number, and the second number, which is in the brackets, represents the section number. For example, the value 1 (6) in row $l = 8$ and column $k = 1$ means that Section 6 of Course 1 was assigned to Room 8 at 9.00–12.00 a.m. on Day 1 for the examination because $x_{1,6,8} = 1$ and $y_{1,1} = 1$. Similarly, the cell in row $l = 17$ and column $k = 1$ means that Sections 1–3 of Course 1 were assigned to Room 17 ($x_{1,1,17} = x_{1,2,17} = x_{1,3,17} = 1$) in the morning on Day 1 ($y_{1,1} = 1$), and the cell in row $l = 2$ and column $k = 1$ means that Sections 4 and 5 of Course 1 were assigned to Room 2 ($x_{1,4,2} = x_{1,5,2} = 1$) at the same Time Slot 1 ($y_{1,1} = 1$) for the examination. Thus, it was possible for more than one section of the same course to be assigned to the same room for the examination. In this example, Course 1 required a minimum of three rooms for the examination in Time Slot 1, that is, Rooms 2, 8, and 17 ($z_{1,1,2} = z_{1,1,8} = z_{1,1,17} = 1$). In Table 8, moreover, an empty cell gives the information that there is no examination in that room at that time slot and day. For example, there is no examination in Room 1 in the morning (9.00–12.00 a.m.) on Day 2 because the cell at row $l = 1$ and column $k = 3$ is empty. The other empty and nonempty cells can be interpreted similarly.

The objective value of the optimal solution for Problem 1 returned from the CPLEX solver is 4.038. This means that the examination schedule of Problem 1 required a minimum of 4 days to cover all the exams ($d = 4$), and the number of rooms required for the examination of all courses over the entire examination schedule is 38 (this value is equal to the number of nonempty cells in Table 8).

4.2.2. Results of Problem 2. In this subsection, Problem 2 was solved using the proposed ILP model. The model was implemented in the ILOG OPL CPLEX 12.6 software running on a computer with an Intel Xeon Platinum 8168 CPU at 2.70 GHz and 32 GB RAM. The model size of Problem 2 is also shown in Table 7. The maximum running time of Problem 2 was limited to 2 h. After solving the model, the result showed that Problem 2 could not be solved to an optimal solution within the 2-h time limit, so we reported the best feasible solution that could be found within this time limit (incumbent solution). The examination schedule derived from the incumbent solution is shown in Table 9.

As shown in Table 9, the midterm examination schedule required a minimum of 5 days to cover all the exams (10 time slots). This value is smaller than the maximum number of available midterm examination days of the CIT (6 days). Note that, according to Table 9, all courses in any curriculum were assigned to different time slots as dictated by Constraint (3). For example, Curriculum 1 has Courses 1, 2, 11, 12, 13, 14, 15, 17, and 22 assigned to Time Slots 2, 6, 8, 3, 5, 9, 7, 10, and 4, respectively ($y_{1,2} = y_{2,6} = y_{11,8} = y_{12,3} = y_{13,5} = y_{14,9} = y_{15,7} = y_{17,10} = y_{22,4} = 1$). This showed that Constraint (3) was satisfied, and an exam conflict does not occur so that no student has to take more than one exam at the same time.

Furthermore, according to Table 9, all sections of a course were assigned to the same time slot. For example, Course 21 has 17 sections assigned to Time Slot 4 (1.00–4.00 p.m. on Day 2), and this course used Rooms 12, 15, 20, 29, 39, 44, 47, 50, 59, and 61 for the examinations (see column $k = 4$). Some examination rooms are used for exams of more than one section. For example, the values in row $l = 15$ and column $k = 4$ mean that Sections 2 and 17 of Course 21 were assigned to Room 15 for the midterm examination. The number of students from these two sections is 80 ($m_{21,2} = 50$ and $m_{21,17} = 30$), which is not greater than the capacity of this room ($c_{15} = 80$). This showed that Constraint (8), the room capacity constraint, in the model was satisfied. Room 61, which is the largest room, was also used for the examination of Sections 3, 5, 10, and 15 of Course 21 (see row $l = 61$ and column $k = 4$). These four sections were scheduled to have their exams in the same room, where the number of all students in this room ($m_{21,3} + m_{21,5} + m_{21,10} + m_{21,15} = 50 + 50 + 46 + 54 = 200$) is not greater than the capacity of this room ($c_{61} = 200$). The other nonempty cells in Table 9 can be interpreted similarly.

Another example of a course examination is described as follows. From Table 4, Course 1 has three sections with 54, 59, and 50 students in each respective section. Additionally, Table 6 indicates that students from only two curriculums take this course: Curriculums 1 and 2. In the first semester of the academic year 2021, Curriculum 1 has a total of 113 students. The students in Curriculum 1 are divided into two sections for Course 1, with Sections 1 and 2 having 54 and 59 students, respectively. Curriculum 2 has a total of 50 students, and these students are in Section 3 of Course 1. According to rows $l = 59$ and $l = 38$ in column $k = 2$ of Table 9, the students from Curriculum 1 have their midterm examinations for Course 1 at 1.00–4.00 p.m. on Day 1, where the students in Sections 1 and 2 will take the examination in Rooms 59 and 38, respectively. Similarly, the students from Curriculum 2 (in Section 3) have their examinations for Course 1 at the same time, but they will take the examination in Room 51 according to row $l = 51$ and column $k = 2$ of Table 9. The examination for other courses can be interpreted in a similar way.

The objective value of the incumbent solution for Problem 2 returned from the CPLEX solver is 5.237. This means that the minimum number of examination days required to cover all the exams (d) for the midterm examination is 5. In addition, the minimum number of rooms

Table 4: The number of courses, sections, and students in each section for Problem 2.

Course $i \in S$	Number of sections of course i (g_i)	Number of students in each section $(m_{i1}, m_{i2}, \ldots, m_{i,g_i})$
1	3	54, 59, 50
2	1	46
3	16	50, 50, 50, 50, 50, 39, 50, 50, 50, 46, 50, 50, 50, 40, 46, 50
4	2	50, 55
5	2	50, 39
6	2	42, 50
7	2	50, 56
8	2	49, 50
9	2	36, 30
10	13	50, 50, 50, 50, 50, 39, 50, 50, 50, 46, 50, 50, 50
11	14	50, 50, 50, 50, 50, 39, 50, 50, 50, 46, 50, 50, 50, 50
12	21	50, 50, 50, 50, 50, 39, 50, 50, 50, 46, 50, 50, 50, 40, 46, 50, 59, 59, 48, 58, 54
13	18	50, 50, 50, 50, 50, 39, 50, 50, 50, 46, 50, 50, 50, 40, 46, 50, 50, 50
14	14	50, 50, 50, 50, 50, 39, 50, 50, 50, 46, 50, 50, 50, 50
15	24	50, 50, 50, 50, 50, 39, 50, 50, 50, 46, 50, 50, 50, 40, 46, 50, 59, 59, 48, 58, 54, 50, 50, 50
16	12	50, 50, 50, 50, 50, 39, 50, 50, 50, 46, 50, 50
17	14	50, 50, 50, 50, 50, 39, 50, 50, 50, 46, 50, 50, 50, 50
18	7	54, 38, 47, 54, 35, 50, 32
19	6	54, 38, 47, 54, 35, 50
20	13	50, 50, 50, 50, 50, 39, 50, 50, 50, 46, 50, 50, 50
21	17	50, 50, 50, 50, 50, 39, 50, 50, 50, 46, 50, 50, 50, 40, 54, 57, 30
22	21	50, 50, 50, 50, 50, 39, 50, 50, 50, 46, 50, 50, 50, 40, 54, 57, 30, 50, 50, 50, 50
23	11	50, 50, 50, 50, 50, 39, 50, 50, 50, 46, 50
24	3	50, 33, 32
25	3	50, 33, 32
26	6	50, 50, 50, 50, 50, 50
27	4	50, 50, 50, 50
28	2	50, 50
29	1	50
30	1	50
31	1	50
32	2	50, 50
33	3	50, 50, 50
34	5	50, 50, 50, 50, 50
35	1	50
36	2	50, 50
37	2	50, 50
38	2	50, 50
39	3	50, 50, 50
40	1	50
41	5	50, 50, 50, 50, 50
42	1	50
43	1	50
44	1	50
45	1	50
46	2	50, 50
47	1	50
48	1	50
49	1	50
50	1	50
51	1	50
52	2	50, 50
53	2	50, 50
54	5	50, 50, 50, 50, 50
55	1	50
56	2	50, 50
57	2	50, 50
58	2	50, 50

Table 4: Continued.

Course $i \in S$	Number of sections of course i (g_i)	Number of students in each section $(m_{i1}, m_{i2}, \ldots, m_{i,g_i})$
59	3	50, 50, 50
60	2	50, 50
61	2	50, 50
62	2	50, 50
63	1	50
64	1	50
65	3	50, 50, 50
66	3	50, 50, 50
67	5	50, 50, 50, 50, 50
68	1	50
69	1	50
70	1	50
71	1	50
72	3	50, 50, 50
73	1	50

required for the examination of all courses over the entire examination schedule is 237, which is equal to the number of nonempty cells in Table 9.

Furthermore, from Table 9, it can be seen that large rooms are used for examinations more frequently than small rooms, especially Rooms 9, 12, and 20–23, which have a capacity of 100, and Room 61, which has a capacity of 200. These rooms are used for examinations in everyday because they can accommodate more than one section of a course to have their exams in the same room. This can reduce the number of times the examination rooms are used compared to assigning each section of a course to a separate examination room, which can increase the utilization of the resources. Using the proposed model, the number of examination rooms used (237 rooms) decreased by 104 compared to assigning each course section to a separate examination room (341 rooms).

5. Discussion

The results from the proposed ILP model gave suitable information on which course section should be assigned to which room, at which time slot on which day for the midterm examination. These are helpful and practical in the real situation. Note that the proposed model is elementarily useful in that it can be used to find whether all the courses provided by the CIT (in Problem 2) can be scheduled within the maximum limit of six available examination days. The results in Table 9 showed that it is feasible, with the minimum number of examination days used over the entire examination period from the incumbent solution being five. Thus, the CIT can use the data in Table 9 to provide the midterm examination schedule and release it a few weeks before the examination period. This will allow each student in every course section to know their exam room and enter it correctly on the examination day.

The results of Problem 2 also provided valuable information that examinations will begin on a Monday and end on the Friday, with no need for Saturday to be used because the number of examination days required to cover all the exams is five. This is preferable for both proctors and students in a real-life situation because they usually want to work only on weekdays and not on weekends. Moreover, the CIT can reduce the cost of utilities on Saturday since there is no examination. In addition, the wage of a proctor who works on a weekend is higher than that on a weekday. Therefore, the CIT can also save on the cost of proctors in weekends.

As for the computational time, the small problem (Problem 1 with 69 course sections) could be solved optimally within a reasonable computational time of 2 min and 46 s, while the large problem (Problem 2 with 341 course sections) could not be solved optimally within a time limit of 2 h. This is because the model size (the number of binary variables and constraints) greatly increases when there are many course sections in the problem, which also increases the complication of the model. It is common to take a long computational time when solving a large-sized ILP problem. However, some practitioners can accept a promise solution within an acceptable time instead of an optimal solution. In practice, therefore, the practitioners can use the proposed ILP model to find a good feasible solution within an acceptable computational time and use it as a good examination schedule in real situation. Nowadays, the administrative staff at the CIT manually schedule the midterm examination timetable for all first-year courses, which takes more than one week to create a feasible examination timetable. Moreover, the manually scheduled examination timetable is often accompanied by exam conflicts, which require much time to resolve. However, the proposed ILP model can find a feasible conflict-free timetable for the midterm examination in just 2 h, which saves much time in creating a feasible examination timetable for the CIT and can greatly reduce the task of the staff handling the timetabling process at the CIT. These show the usefulness of the proposed ILP model.

The proposed ILP model can also be further extended to be more practical for real-life applications. For example, if there are some students at the CIT who are not following their curriculums, such as second-, third-, or fourth-year

Table 5: The capacity of each room for Problem 2.

Room number ($l \in R$)	Capacity (c_l)
1	50
2	50
3	50
4	50
5	50
6	50
7	50
8	50
9	100
10	50
11	50
12	100
13	50
14	70
15	80
16	60
17	60
18	70
19	60
20	100
21	100
22	100
23	100
24	80
25	50
26	50
27	50
28	50
29	50
30	50
31	50
32	50
33	70
34	70
35	70
36	80
37	50
38	70
39	50
40	80
41	50
42	50
43	50
44	50
45	50
46	50
47	50
48	50
49	50
50	80
51	50
52	50
53	50
54	50
55	50
56	50
57	60
58	60
59	60
60	50
61	200
62	50
63	50
64	50
65	50
66	50

Table 6: List of courses in each curriculum for Problem 2.

Curriculum $h \in N$	Courses in curriculum h (A_h)
1	{1, 2, 11, 12, 13, 14, 15, 17, 22}
2	{1, 11, 12, 13, 14, 15, 16, 17, 22}
3	{3, 4, 11, 12, 13, 14, 15, 17}
4	{3, 10, 13, 15, 16, 17, 21, 25}
5	{3, 10, 11, 12, 13, 14, 15, 19, 22}
6	{9, 11, 12, 13, 14, 15, 16, 17, 21}
7	{3, 5, 11, 12, 13, 14, 15, 17, 22}
8	{3, 7, 8, 13, 14, 15, 17, 21}
9	{6, 11, 12, 13, 14, 15, 17, 24}
10	{10, 13, 14, 15, 16, 17, 18, 20, 22}
11	{10, 13, 14, 15, 16, 17, 18, 20, 22}
12	{10, 13, 14, 15, 16, 17, 18, 20, 22}
13	{3, 11, 12, 13, 14, 15, 16, 17}
14	{3, 11, 12, 13, 14, 15, 16, 17, 21}
15	{3, 10, 13, 14, 15, 16, 17, 22}
16	{3, 11, 12, 13, 14, 15, 16, 17}
17	{3, 13, 14, 15, 16, 17, 18, 20, 22}
18	{10, 11, 12, 13, 14, 15, 17, 21, 23}
19	{10, 11, 12, 13, 14, 15, 17, 21, 23}
20	{26, 27, 28, 29, 30}
21	{26, 27, 28, 29, 31}
22	{32, 33, 34, 35, 36, 37, 38, 39}
23	{26, 27, 29, 30, 31}
24	{41, 42, 45, 46, 47, 48}
25	{40, 41, 42, 43, 44, 45}
26	{41, 49, 50, 51, 52, 53, 54, 55}
27	{27, 56, 57, 58, 59, 60, 61}
28	{27, 62, 63, 64, 65, 66, 67}
29	{27, 65, 66, 67, 68, 69, 70}
30	{27, 63, 65, 66, 67, 71, 72, 73}

Table 7: Model size of Problems 1 and 2.

Type	Problem 1	Problem 2
Binary variable	8085	81,198
Integer variable	1	1
Constraint	16,578	299,838

students taking first-year courses in the current semester, we can add additional constraints to deal with it as follows. Suppose that Student 1, who is a second-year student, enrolls in first-year courses, which consist of Courses 11, 13, and 14 in this semester. In addition, Student 1 also enrolls in second-year courses, which consist of Courses 74, 75, and 76 in this semester (we assumed that the second-year courses

Table 8: The examination schedule derived from an optimal solution of Problem 1.

Room (l)	Time slot							
	Day 1		Day 2		Day 3		Day 4	
	$k=1$ (Mor)	$k=2$ (Aft)	$k=3$ (Mor)	$k=4$ (Aft)	$k=5$ (Mor)	$k=6$ (Aft)	$k=7$ (Mor)	$k=8$ (Aft)
1	17 (1), 17 (2)	20 (1), 20 (2)					16 (1), 16 (2)	
2	1 (4), 1 (5)					21 (1), 21 (2)		15 (1), 15 (2)
3	10 (1)	3 (1), 3 (2)		19 (1), 19 (2)		12 (1), 12 (2)		24 (1), 24 (3)
4	27 (1)		11 (1), 11 (2)				7 (1), 7 (2)	
5								24 (2)
6				14 (1)		6 (1)	22 (1)	
7	23 (1)							
8	1 (6)				4 (1)		28 (1)	
9	30 (1)							
10						32 (6)		
11								
12		8 (1)		5 (1)	9 (1)			
13						29 (1)		
14								
15					26 (3)			
16								13 (1), 13 (2)
17	1 (1), 1 (2), 1 (3)	31 (1), 31 (2), 31 (3)	2 (1), 2 (2), 2 (3), 2 (4)	25 (1), 25 (2), 25 (3)	26 (1), 26 (2), 26 (4), 26 (5)	18 (1), 18 (2), 18 (3)	7 (3), 7 (4), 7 (5)	13 (3), 13 (4), 13 (5)

Note: "Mor" represents 9.00 a.m.–12.00 a.m. (Morning) and "Aft" represents 1.00 p.m.–4.00 p.m. (Afternoon).

Table 9: The midterm examination schedule derived from the incumbent solution of Problem 2.

Room (l)	Time slot									
	Day 1		Day 2		Day 3		Day 4		Day 5	
	$k=1$ (Mor)	$k=2$ (Aft)	$k=3$ (Mor)	$k=4$ (Aft)	$k=5$ (Mor)	$k=6$ (Aft)	$k=7$ (Mor)	$k=8$ (Aft)	$k=9$ (Mor)	$k=10$ (Aft)
1	54 (2)	3 (3)	59 (1)		64 (1)	50 (1)	15 (7)	41 (1)		34 (4)
2				22 (10)				20 (7)		
3	19 (6)	23 (6)		39 (1)	72 (1)		60 (2)	11 (7)		
4		32 (2)		22 (5)		61 (2)		11 (10)		
5				69 (1)						
6	54 (5)	68 (1)				2 (1)				
7	27 (2)		12 (4)	22 (2)		65 (3)		20 (6)		
8		23 (1)								
9	16 (7), 16 (8)	3 (2), 3 (13)	12 (7), 12 (19)	39 (2), 39 (3)	13 (5), 13 (17)	10 (5), 10 (12)	15 (16), 15 (24)	11 (6), 11 (13)	57 (1), 57 (2)	17 (7), 17 (11)
10		3 (10)	45 (1)							
11	19 (3)									
12	16 (4), 16 (11)	26 (1), 26 (5)	33 (1), 33 (2)	21 (4), 21 (8)	13 (15), 13 (18)	28 (1), 28 (2)	15 (3), 15 (11)	11 (1), 11 (4)	14 (3), 14 (11)	17 (12), 17 (13)
13		55 (1)		48 (4)	35 (1)		44 (7)	41 (5)	25 (1)	40 (1)
14	7 (2)	23 (5)	59 (3)				15 (17)			
15	24 (2), 24 (3)	3 (8)	18 (2), 18 (5)	21 (2), 21 (17)			15 (13)	66 (2)		17 (9)
16			63 (1)	22 (15)			15 (18)			
17	16 (2)	3 (12)	12 (18)				15 (5)			
18			18 (6)	22 (9)				20 (13)		
19	19 (1)		12 (20)						14 (8)	67 (2)
20	36 (1), 36 (2)	3 (1), 3 (15)	12 (2), 12 (11)	21 (11), 21 (13)	13 (8), 13 (9)	10 (7), 10 (10)	37 (1), 37 (2)	11 (5), 11 (11)	38 (1), 38 (2)	67 (1), 67 (5)
21	16 (5), 16 (9)	23 (4), 23 (11)	12 (8), 12 (10)	22 (8), 22 (21)	13 (1), 13 (16)	65 (1), 65 (2)	15 (22), 15 (23)	11 (3), 11 (8)	14 (5), 14 (6)	17 (2), 17 (8)
22	27 (1), 27 (3)	23 (7), 23 (10)	12 (5), 12 (13)	22 (4), 22 (13)	72 (2), 72 (3)	10 (1), 10 (8)	15 (4), 15 (12)	20 (1), 20 (3)	14 (4), 14 (14)	34 (1), 34 (3)
23	54 (1), 54 (4)	3 (4), 3 (9)	12 (12), 12 (16)	22 (11), 22 (12)	13 (12), 13 (10)	10 (9), 10 (11)	15 (2), 15 (9)	41 (2), 41 (4)	14 (2), 14 (12)	17 (10), 17 (14)
24					13 (6), 13 (14)		15 (1)	20 (2)		
25	4 (1)									
26	27 (4)		53 (2)		58 (1)					
27	16 (1)		59 (2)							49 (10)
28								11 (14)	14 (1)	
29			12 (1)	21 (9)				66 (3)	51 (9)	34 (2)
30				29 (1)						
31	5 (1)	3 (5)					60 (1)			
32										
33			18 (1)	22 (16)	70 (1)		15 (20)			
34	4 (2)						62 (2)			
35		3 (16)	12 (17)				46 (2)			67 (3)
36		3 (7)		22 (17), 22 (18)	30 (1)	61 (1)		66 (1)		
37			8 (1)	22 (7)			62 (1)			
38		1 (2)								17 (5)
39				21 (12)				20 (10)		
40	19 (2), 19 (5)	3 (6), 3 (14)	12 (6), 12 (14)	22 (6), 22 (14)			15 (21)		25 (2), 25 (3)	
41		43 (1)			52 (2)					34 (5)

Table 9: Continued.

Room (l)	Time slot									
	Day 1		Day 2		Day 3		Day 4		Day 5	
	$k=1$ (Mor)	$k=2$ (Aft)	$k=3$ (Mor)	$k=4$ (Aft)	$k=5$ (Mor)	$k=6$ (Aft)	$k=7$ (Mor)	$k=8$ (Aft)	$k=9$ (Mor)	$k=10$ (Aft)
42			53 (1)	22 (1)	52 (1)					
43		56 (2)	33 (3)				46 (1)			
44		23 (9)		21 (1)						
45	54 (3)	71 (1)						11 (2)		
46								41 (3)		
47	7 (1)	3 (11)		21 (7)						
48			8 (2)		58 (2)					
49		56 (1)								
50	24 (1)		18 (3), 18 (7)	21 (6), 21 (14)		9 (1), 9 (2)	15 (6), 15 (14)	31 (1)		
51		1 (3)			42 (1)					
52								20 (4)		
53				73 (1)						
54						6 (2)				
55				22 (19)		6 (1)				
56		23 (2)								
57			18 (4)							
58								20 (9)		
59	19 (4)	1 (1)		21 (16)						
60								11 (12)		
61	16 (3), 16 (6), 16 (10), 16 (12)	26 (2), 26 (3), 26 (4), 26 (6)	12 (3), 12 (9), 12 (15), 12 (21)	21 (3), 21 (5), 21 (10), 21 (15)	13 (3), 13 (7), 13 (11), 13 (13)	10 (2), 10 (3), 10 (4), 10 (6)	15 (8), 15 (10), 15 (15), 15 (19)	20 (5), 20 (8), 20 (11), 20 (12)	14 (7), 14 (9), 14 (10), 14 (13)	17 (1), 17 (3), 17 (4), 17 (6)
62		23 (3)								
63		23 (8)								
64	5 (2)			22 (20)	13 (2)				47 (1)	67 (4)
65		32 (1)				10 (13)				
66				22 (3)	13 (4)			11 (9)		

have course indexes that continue from the first-year courses in Table 4). Therefore, the set of all courses that must be studied by Student 1 (B_1) in this semester is $B_1 = \{11, 13, 14, 74, 75, 76\}$. To avoid exam conflicts, all six courses in B_1 cannot be scheduled to have their exams at the same time. Thus, we can add the constraint $\sum_{i \in B_1} y_{ik} \leq 1$, $\forall k \in T$ to the proposed ILP model. In general, suppose that there are p students who are not following the curriculum. Let B_h ($h \in P = \{1, 2, \ldots, p\}$) be the set of all courses that must be studied by Student $h \in P$ in the current semester. Then, we can add the constraint $\sum_{i \in B_h} y_{ik} \leq 1, \forall k \in T, \forall h \in P$ to the proposed model to avoid exam conflicts for all students. The role of this constraint is similar to Constraint (3) in the proposed model.

6. Conclusion

This paper reports an ETTP using the case study of the CIT. The goal was to develop a new ILP model that could automatically generate an examination timetable for the CIT. This is a new application for the CIT because an ILP method for solving the ETTP at the CIT has not been investigated in the literature. The objective of the proposed ILP model was to minimize the number of examination days required to cover all the exams and to minimize the number of rooms required for the examinations of all courses, so as to increase the utilization of the resources of the CIT. The constraints in the proposed model included avoiding exam conflicts, allowing an examination room to accommodate exams from more than one course section, ensuring that the examinations for all course sections occur simultaneously, and permitting each examination room to have the exam of only one course at a time.

To evaluate the performance of the proposed ILP model, the real-world data from the CIT were applied. These data consisted of 73 courses with 341 sections and 66 available rooms. The proposed model was used to solve two test problems. The first problem (small problem) was generated using partial data from the entire data, which included 32 courses with 69 sections and 17 available rooms. The second problem (large problem) used the whole data from the CIT to create a timetable for the midterm examination under the requirements at the CIT, including a maximum midterm examination period of 6 days. All problems were solved using the CPLEX solver. The results showed that the first problem could be solved to reach optimality with a computational time of only 2 min and 46 s, yielding an optimal examination timetable. For the second problem, an optimal solution could not be achieved within a 2-h time limit due to the large model size, but a good feasible examination schedule without any conflicts could be achieved within the 2-h time limit. This is useful and practical for the administrative staff at the CIT since they could accept a good feasible examination schedule within an acceptable time in a real situation, and it saves much time compared to manual scheduling by the administrative staff. The obtained examination schedule from the proposed model satisfied all the requirements of the CIT and also required only five examination days to cover all the exams. The proposed model is valuable to planners or decision-makers who want to find a good examination timetable at the CIT. It also illustrated the application of ILP to handle the problem from the real world and can be used as an option to provide an optimized schedule at the CIT. It may also be a potential model for other institutes or may be applicable to scheduling an examination timetable at other universities if their constraints are similar to the constraints of the CIT.

For future research, the proposed model could be extended or modified for other particular rules. Some additional constraints can be introduced to the problem, such as spreading the exams throughout the examination period, combining some small exams of different courses in the same examination room, or splitting some large exams into several rooms. A higher priority for some exams to be scheduled before other exams and the introduction of room types to satisfy special requirements of some exams (e.g., an exam needs to use a laboratory room or a computer room or have special needs access) are also interesting additional constraints. Furthermore, solving the problem using data with a larger size than this research study is also interesting. For example, extending the problem to schedule the examination for the overall courses of all bachelor's degree students at the CIT. In addition, developing a heuristic or a metaheuristic approach for solving the problem would also be a very challenging but required task when the size of the problem is very large.

The limitation of this paper is that it is assumed that each course at the CIT requires the same duration for the examination. In reality, however, different courses can require a different time duration for their examinations, which cannot be solved using the proposed model.

Data Availability Statement

The data used to support the findings of this study are already included in the article.

Conflicts of Interest

The authors declare no conflicts of interest.

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